Cosmological Perturbation Theory

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Abstract

In this report, we have started with the discussion of homogeneity and isotropy and used them to say that spatial homogeneity and isotropy of a space leads it to be maximally symmetric and have it's metric of the form of FRW. Then we touch up a bit on the perfect fluid and how we get the equation for it's energy-momentum tensor. After this we have for the FRW metric stated the different curvature quantities and the Einstein's and the Conservation equations. After discussing about fully homogeneous and isotropic spatial space we stress upon the need for perturbations to define our universe and ahead in the report we introduce Linear perturbations to the background metric and the energy-momentum tensor. We then stress upon SVT decomposition of these perturbations and solve for vector and scalar modes. For vector modes, we write the Einstein's and conservations equation generated from the code and write them in terms of the guage invariant quantities and finally solve them and prove that vector modes decay. For scalar modes we write the Einstein's and conservation equations and also find the gauge invariant quantities for this mode.

We have used SageMath to write code that generates various perturbed/unperturbed curvature quantities, Einstein's and Conservation equations for a given spacetime metric.

Github Repo: Cosmological Perturbation Theory Repository

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1 The FRW Spacetime Metric

1.1 Homogeneity and Isotropy

Contemporary models of the universe are based on an idea called the **Copernican Principle**, which states that the universe is mostly the same everywhere. Which seems absurd, but because we apply this principle at large scales, we can average the local variations and it is pretty much consistent with observations. The principle is based on 2 very important mathematical properties **Homogeneity** and **Isotropy**.

Homogeneity means that the geometry defined by a metric is identical at every point via isometries which are transformations that preserve a metric, formally they are a map $\phi : M \to M$ such that $\phi^* g = g$ (ϕ^* is the pullback), M being the manifold on which the metric g is defined. In other words, given any 2 points p and q, there exists an isometry that takes p into q.

Isotropy at a point p means that no matter where we look at from p, the space looks the same. In other words, a manifold M is isotropic around p if for any 2 vectors **V** and **W** in T_pM(the tangent plane at p on M) there exists an isotropy of M such that the pushforward (ϕ_*) of **W** under that isometry is parallel to **V**.

1.2 Maximally Symmetric Spaces

(*Refer to Appendix A to understand killing vectors).

Homogeneity and Isotropy of an n-dimensional space lead it to have the maximum possible number of killing vectors which will be:

$$\frac{n(n+1)}{2} \tag{1}$$

i.e. the maximum number of solutions of the killing's equation. Now, the n-dimensional space $\mathbf{R}^n$ with a flat euclidean metric achieves maximum number of solutions, hence we can solve for this space and the formula will be valid for any n-dimensional space. In this space we will have n axes. Hence, for translations we have **n** killing vectors. Now, we have n axes which can be rotated to any n-1 axes. Hence, we have $\mathbf{^nC_2} = \frac{\mathbf{n(n-1)}}{\mathbf{2}}$ killing vectors corresponding to rotations. So in total we have the maximum possible number of killing vectors for an n-dimensional manifold to be:

$$n + \frac{n(n-1)}{2} = \frac{n(n+1)}{2}$$

A space with the maximum number of killing vectors possible is called a
Maximally Symmetric Space.

For a maximally symmetric space the curvature (Ricci Scalar R) shall be the same at every point and in every direction. Therefore, if we know the curvature at one particular point we know the curvature at every point in the space.

We can construct the Riemann tensor components from this scalar. We can go to an infinitesimal part of the space and choose a locally inertial set of coordinates at a point p hence $\hat{g}_{\mu\nu} = \hat{\eta}_{\mu\nu}$. Now these set of coordinates are not unique cause we can perform a Lorentz transformation at p and still the components of the metric will be those of $\hat{\eta}_{\mu\nu}$. Similarly, because the space is maximally symmetric we want Riemann tensor components to also not change under Lorentz transformations. Now, there are some unique tensors that do not change under Lorentz transformations eg: the metric, Kronecker delta and the Levi-Civita tensor. Hence, the Riemann tensor will be proportional to these invariant tensors.

There is one unique combination that matches the Riemann tensor's symmetries:

$$R_{\hat{\rho}\hat{\sigma}\hat{\mu}\hat{\nu}} \ \alpha \ g_{\hat{\rho}\hat{\mu}}g_{\hat{\sigma}\hat{\nu}} - g_{\hat{\rho}\hat{\nu}}g_{\hat{\sigma}\hat{\mu}}$$

We can find the proportionality constant k by contracting the LHS and RHS:

$$\begin{aligned}
&R_{\hat{\rho}\hat{\sigma}\hat{\mu}\hat{\nu}} = k(g_{\hat{\rho}\hat{\mu}}g_{\hat{\sigma}\hat{\nu}} - g_{\hat{\rho}\hat{\nu}}g_{\hat{\sigma}\hat{\mu}}) \\
&\Rightarrow g^{\hat{\rho}\hat{\mu}}R_{\hat{\rho}\hat{\sigma}\hat{\mu}\hat{\nu}} = k(g^{\hat{\rho}\hat{\mu}}g_{\hat{\rho}\hat{\mu}}g_{\hat{\sigma}\hat{\nu}} - g^{\hat{\rho}\hat{\mu}}g_{\hat{\rho}\hat{\nu}}g_{\hat{\sigma}\hat{\mu}}) \\
&\Rightarrow R_{\hat{\sigma}\hat{\nu}} = k(ng_{\hat{\sigma}\hat{\nu}} - g_{\hat{\sigma}\hat{\nu}}) \\
&\Rightarrow g^{\hat{\sigma}\hat{\nu}}R_{\hat{\sigma}\hat{\nu}} = kg^{\hat{\sigma}\hat{\nu}}g_{\hat{\sigma}\hat{\nu}}(n-1) = k\delta^{\hat{\sigma}}_{\hat{\sigma}}(n-1) \\
&\Rightarrow R = kn(n-1) \\
&\Rightarrow k = \frac{R}{n(n-1)}
\end{aligned}$$

Therefore, our Riemann tensor for a maximally symmetric space is:

$$R_{\hat{\rho}\hat{\sigma}\hat{\mu}\hat{\nu}} = \frac{R}{n(n-1)}(g_{\hat{\rho}\hat{\mu}}g_{\hat{\sigma}\hat{\nu}} - g_{\hat{\rho}\hat{\nu}}g_{\hat{\sigma}\hat{\mu}}) \tag{2}$$

Here, the implication of a maximally symmetric space to have such a Riemann tensor is bidirectional.

1.3 The FRW Metric

The Copernican principle implies the symmetry in both space and time. But through observations it has been observed that our universe is homogeneous and isotropic only spatially and rather evolves with time. Hence, we can write our spacetime to be $\mathbf{R}\times\Sigma$ where $\mathbf{R}$ represents the time direction and Σ represents the spatially homogeneous and isotropic 3-manifold.

We can thus represent our spacetime metric as:

$$ds^2 = -dt^2 + R^2(t)d\sigma^2$$

Here, t is the timelike coordinate, R(t) is a function of t known as the scale factor and should not be confused with the notation for Ricci scalar "R" and $d\sigma^2$ is our infinitesimal distance defined on the Σ manifold, which can be written as:

$$d\sigma^2 = \gamma_{ij}(u)du^i du^j$$

where, (u^1, u^2, u^3) are our coordinates on Σ and γ_{ij} is a maximally symmetric 3-dimensional metric. The scale factor $R(t)$ tells us how big our spatial slice Σ is at time t. Here, we will notice that there are no cross terms like $dt\,du^i$ because homogeneity and isotropy forbid the existence of a preferred spatial direction and the comoving frame is the coordinate system where there is no such preferred direction.

We know that for a maximally symmetric space our Riemann tensor will look like as given in equation (2). Now, considering our metric γ_{ij} which is 3-dimensional and maximally symmetric, the tensor will be:

$$R_{\hat{i}\hat{j}\hat{k}\hat{l}} = \frac{R}{6}(\gamma_{\hat{i}\hat{k}}\gamma_{\hat{j}\hat{l}} - \gamma_{\hat{i}\hat{l}}\gamma_{\hat{j}\hat{k}}) \tag{3}$$

Contracting with γ^{ik} we can get the Ricci tensor:

$$R_{jl} = \frac{R}{6}\gamma_{jl}(3-1) = \frac{R}{3}\gamma_{jl} \tag{4}$$

Now, since we have a constant curvature. Starting from the most general spherically symmetric metric:

$$d\sigma^2 = \gamma_{ij}(u)du^i du^j = e^{2\beta(\bar{r})}d\bar{r}^2 + \bar{r}^2 d\Omega^2$$

Where, $\bar{r}$ is our radial coordinate and $d\Omega^2 = d\theta^2 + sin^2(\theta)d\phi^2$ is our distance on a 2-sphere. By, imposing the above constant curvature condition, our Ricci tensor will be of the form as in equation (4). Solving this equation's rr and $\theta\theta$ component by inserting the values of $g_{rr}, g_{\theta\theta}, R_{rr}, R_{\theta\theta}$, we can find the value of β to be:

$$e^{2\beta(\bar{r})} = \frac{1}{1-k\bar{r}^2} \Rightarrow \beta = -\frac{1}{2}ln(1-k\bar{r}^2)$$

hence, our $d\sigma^2$ on the three-surface Σ shall be:

$$d\sigma^2 = \frac{d\bar{r}^2}{1-k\bar{r}^2} + \bar{r}^2 d\Omega^2$$

We can see in equation (4) that $k = \frac{R}{6}$ and that the value of k can set the curvature R. We normalize k to have the following values:

$$k \in \{+1, 0, -1\}$$

Here k = -1 corresponds to a constant negative curvature on Σ and we call this space **open**, k = 0 means no curvature and we call this space **flat** and k = 1 means a constant positive curvature and we call this space **close**.

We can interpret these cases physically much better by changing the variable $\bar{r}$ to χ defined by:

$$d\chi = \frac{d\bar{r}}{\sqrt{1-k\bar{r}^2}} \Rightarrow \bar{r} = S_k(\chi)$$

here, $S_k(\chi)$ is the function:

$$S_k(\chi) = \begin{cases} sin(\chi) & , k = +1 \\ \chi & , k = 0 \\ sinh(\chi) & , k = -1 \end{cases}$$

Therefore, for k = 0:

$$d\sigma^2 = d\chi^2 + \chi^2 d\Omega^2 = dx^2 + dy^2 + dz^2$$

which is our flat euclidean space. For k = +1 we have:

$$d\sigma^2 = d\chi^2 + sin^2(\chi) d\Omega^2$$

which represents our 3-sphere. For k = -1 we have:

$$d\sigma^2 = d\chi^2 + sinh^2(\chi) d\Omega^2$$

This represents the 3-dimensional space of constant negative curvature, a generalization of a hyperboloid. By bringing in the time part we get our whole metric as:

$$ds^2 = -dt^2 + R^2(t) \left[\frac{d\bar{r}^2}{1 - k\bar{r}^2} + \bar{r}^2 d\Omega^2 \right] \tag{5}$$

This metric is called the **Robertson-Walker (RW) metric**, using the Einstein's equation we can determine the behaviour of R(t).

We can see that the following substitutions:

$$\begin{aligned} &R \to \lambda^{-1} R \\ &\bar{r} \to \lambda \bar{r} \\ &k \to \lambda^{-2} k \end{aligned}$$

will leave equation (5) invariant. Now, in our metric $\bar{r}, \chi$ are actually dimensionless, the scale factor R(t) has units of distance so we can make the following substitutions:

$$\begin{aligned} &R \to a = R_0^{-1} R \\ &\bar{r} \to r = R_0 \bar{r} \\ &k \to R_0^{-2} k \end{aligned}$$

and we will get our metric as:

$$ds^2 = -dt^2 + a^2(t) \left[\frac{dr^2}{1 - kr^2} + r^2 d\Omega^2 \right] \tag{6}$$

Now, r has dimensions of length and a(t), our scale factor is dimensionless. The metric is known as the **FRW (Friedmann-Robertson-Walker)** metric, and for a maximally symmetric space the metric shall be of this form.

2 The Perfect Fluid

The rotational and translational invariance of the energy momentum tensor $T_{\mu\nu}$ will force it to have the form of that of a perfect fluid i.e. $T_{\mu\nu}$ will be:

$$T_{\mu\nu} = p\, g_{\mu\nu} + (p + \rho)\, u_\mu u_\nu \tag{7}$$

Proof: Consider a gas with homogeneous and isotropic distribution of velocities, i.e.

$$f(\vec{v}) = f(|\vec{v}|)$$

(independent of position) and direction-invariant.

Isotropic variation of the velocities can only be found in the comoving frame of the fluid why? because in this frame $u^\mu = \{1, 0, 0, 0\}$ i.e. you are moving with the particles along their world line hence no preferred direction, if we had a boost along the x-axis than all the particles would have a preferred direction along this axis.

(*For more detail in the following section refer §5.1 to §5.5 from Gravitation by Charles W.Misner, Kip S.Thorne and John Archibald Wheeler)
We know that a energy momentum tensor $T_{\mu\nu}$ has the following form:

$$T_{ij} = \sum_A N_{(A)} p_i^{(A)} v_j^{(A)}$$

Here, "A" is a label for a group of particles with the same properties,
$N_{(A)}$ is the number density in that group,
$p_i^{(A)}$ is the momentum of the ith particle in this group and
$v_j^{(A)}$ is the velocity of the jth particle in this group.

Now, since we have an isotropic distribution therefore for any velocity pair (v_i, v_j) there will be a pair $(v_i, -v_j)$ therefore $T_{ij} = 0$ hence the off-diagonal spatial terms shall be zero.

at the same time the diagonal terms like T_{ii} will be:

$$T_{ii} = \sum_A \frac{N_{(A)} m_{(A)} v_i^2}{1 - \beta_{(A)}^2} \tag{8}$$

Now, according to isotropy no direction is unique, hence $T_{xx} = T_{yy} = T_{zz}$ therefore

$$T_{xx} = T_{yy} = T_{zz} = \sum_A \frac{N_{(A)} m_{(A)} v_x^2}{1 - \beta_{(A)}^2}$$

Now, look closely T_{xx} is the product of particles per unit volume multiplied by the velocity in x-direction multiplied by the momentum in the x-direction i.e. T_{xx} is the flux of x-momentum

along a surface perpendicular to the x-direction and pressure p is defined as momentum per unit area per unit time. Therefore, $T_{xx} = p$ therefore overall

$$T_{ij} = \begin{pmatrix} p & 0 & 0 \\ 0 & p & 0 \\ 0 & 0 & p \end{pmatrix}$$

Now, lets see the T_{00} component. T_{00} is the flux of energy across a surface of constant time, a surface of constant time is just space therefore T_{00} is the energy density ρ. Therefore,

$$T_{\mu\nu} = \begin{pmatrix} \rho & 0 & 0 & 0 \\ 0 & p & 0 & 0 \\ 0 & 0 & p & 0 \\ 0 & 0 & 0 & p \end{pmatrix}$$

Now, this tensor can be rewritten as:

$$T_{\mu\nu} = \begin{pmatrix} \rho & 0 & 0 & 0 \\ 0 & p & 0 & 0 \\ 0 & 0 & p & 0 \\ 0 & 0 & 0 & p \end{pmatrix} = \begin{pmatrix} \rho & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} + \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & p & 0 & 0 \\ 0 & 0 & p & 0 \\ 0 & 0 & 0 & p \end{pmatrix} = \rho u_\mu u_\nu + p(\eta_{\mu\nu} + u_\mu u_\nu)$$

hence,

$$T_{\mu\nu} = p\eta_{\mu\nu} + (\rho + p)\, u_\mu u_\nu$$

i.e.

$$T_{\mu\nu} = pg_{\mu\nu} + (\rho + p)\, u_\mu u_\nu \tag{9}$$

3 FRW Einstein and Conservation Equations

3.1 Different Curvature Quantities for the Metric

We have seen above that the homogeneity and isotropy or our space forces our spacetime metric to be the FRW one:

$$ds^2 = -dt^2 + a^2(t)\left[\frac{dr^2}{1-kr^2} + r^2 d\Omega^2\right]$$

For this metric we can calculate the Christoffel Coefficients of the second kind $\Gamma^\lambda_{\mu\nu}$ by the formula:

$$\Gamma^\lambda_{\mu\nu} = \frac{1}{2} g^{\lambda\alpha}\left(\partial_\mu g_{\alpha\nu} + \partial_\nu g_{\mu\alpha} - \partial_\alpha g_{\mu\nu}\right)$$

which would yield us the surviving values:

$$\begin{aligned}
&\Gamma^0{}_{11} = \frac{a\dot{a}}{1-kr^2}, && \Gamma^1{}_{11} = \frac{kr}{1-kr^2},\\
&\Gamma^0{}_{22} = a\dot{a}r^2, && \Gamma^0{}_{33} = a\dot{a}r^2\sin^2\theta,\\
&\Gamma^1{}_{01} = \Gamma^2{}_{02} = \frac{\dot{a}}{a}, && \Gamma^3{}_{03} = \frac{\dot{a}}{a},\\
&\Gamma^1{}_{22} = -r(1-kr^2), && \Gamma^1{}_{33} = -r(1-kr^2)\sin^2\theta,\\
&\Gamma^2{}_{12} = \Gamma^3{}_{13} = \frac{1}{r}, &&\\
&\Gamma^2{}_{33} = -\sin\theta\cos\theta, && \Gamma^3{}_{23} = \cot\theta.
\end{aligned}$$

Now, the Ricci Tensor values $R_{\mu\nu}$ can be found by the formula:

$$R_{\mu\nu} = \partial_\alpha \Gamma^\alpha_{\mu\nu} - \partial_\nu \Gamma^\alpha_{\mu\alpha} + \Gamma^\alpha_{\alpha\sigma}\Gamma^\sigma_{\mu\nu} - \Gamma^\alpha_{\mu\sigma}\Gamma^\sigma_{\nu\alpha}$$

which shall yield us the surviving values:

$$\begin{aligned}
R_{00} &= -3\frac{\ddot{a}}{a},\\
R_{11} &= \frac{a\ddot{a} + 2\dot{a}^2 + 2k}{1-kr^2},\\
R_{22} &= \left(a\ddot{a} + 2\dot{a}^2 + 2k\right) r^2,\\
R_{33} &= \left(a\ddot{a} + 2\dot{a}^2 + 2k\right) r^2\sin^2\theta
\end{aligned}$$

We can find the Ricci Scalar $R = g^{\mu\nu}R_{\mu\nu}$ as:

$$R = \frac{6\left(\dot{a}^2 + k + a\ddot{a}\right)}{a^2}$$

The $T_{\mu\nu}$ values in the coming frame ($u_\mu = \delta^0_\mu$, $u_\nu = \delta^0_\nu$) can be given by the formula:

$$T_{\mu\nu} = (\rho + p)\delta^0_\mu\delta^0_\nu + p g_{\mu\nu}$$

and hence the surviving values will be:

$$\begin{aligned}
T_{00} &= \rho \Rightarrow T^0_0 = -\rho \\
T_{11} &= pg_{11} \Rightarrow T^1_1 = g^{11}p \\
T_{22} &= pg_{22} \Rightarrow T^2_2 = g^{22}p \\
T_{33} &= pg_{33} \Rightarrow T^3_3 = g^{33}p
\end{aligned}$$

and T will be:

$$T = -\rho + 3p$$

3.2 Einstein's Equations

Now, we can use the Einstein's equation which can be manipulated in the following form:

$$\begin{aligned}
& R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R = -8\pi G T_{\mu\nu} \\
\Rightarrow\ & g^{\mu\nu}R_{\mu\nu} - g^{\mu\nu}\frac{1}{2}g_{\mu\nu}R = -8\pi G g^{\mu\nu}T_{\mu\nu} \\
\Rightarrow\ & R - 2R - -8\pi G T \\
\Rightarrow\ & R = 8\pi G T \\
\Rightarrow\ & R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}(8\pi G T) = -8\pi G T_{\mu\nu} \\
\Rightarrow\ & R_{\mu\nu} = -8\pi G(T_{\mu\nu} - 1/2 g_{\mu\nu}T)
\end{aligned} \tag{10}$$

Putting $\mu = 0, \nu = 0$ shall give us:

$$\frac{\ddot{a}}{a} = \frac{4\pi G(\rho + 3p)}{3} \quad \text{(Raychaudari Equation)} \tag{11}$$

Putting $\mu = 1, \nu = 1$ shall give us:

$$\begin{aligned}
& a\ddot{a} + 2\dot{a}^2 + 2k = -4\pi G a^2(\rho - p) \\
\Rightarrow\ & \frac{\ddot{a}}{a} + 2\frac{\dot{a}^2}{a^2} + \frac{2k}{a^2} = -4\pi G(\rho - p) \\
\Rightarrow\ & \frac{2(\dot{a}^2 + k)}{a^2} = 4\pi G(\rho - p + \rho/3 + p) = \frac{16\pi G\rho}{3}
\end{aligned}$$

$$\Rightarrow \frac{\dot{a}^2}{a^2} = H^2 = \frac{8\pi G\rho}{3} - \frac{k}{a^2} \quad \text{(Friedmann Equation)} \tag{12}$$

Putting $\mu = 2, 3$ and $\nu = 2, 3$ shall also give us the Friedmann Equation.

3.3 Conservation Equations

Now, Let's see the conservation equation $\nabla_\mu T^\mu_\nu = 0$.
Putting $\nu = 0$ (Energy Conservation), we'll get:

$$\dot{\rho} + \frac{3\dot{a}}{a}(\rho + p) = 0 \tag{13}$$

Putting $\nu = j$ (Momentum Conservation), we'll get:

$$\partial_j p = 0 \tag{14}$$

4 Need For Perturbations

In the above calculations we have assumed an isotropic and a homogeneous space. But, we know that in our universe stuff like galaxies, galaxy clusters, us human beings exist which means the space is not homogeneous and isotropic.

We'll assume that throughout the history of our universe the fluctuations from homogeneity and isotropy have been small and hence we shall treat these fluctuations as first-order perturbations. The observable universe is nearly homogeneous and it's spatial curvature is very small, we will take the unperturbed metric to have the FRW form with curvature constant k = 0. Hence, our total metric will be:

$$g_{\mu\nu} = \bar{g}_{\mu\nu} + h_{\mu\nu} \tag{15}$$

Where $\bar{g}_{\mu\nu}$ is the unperturbed flat FRW. In Cartesian coordinate system then it's components will be:

$$\bar{g}_{00} = -1, \quad \bar{g}_{i0} = \bar{g}_{0i} = 0, \quad \bar{g}_{ij} = a^2(t)\delta_{ij}$$

and $h_{\mu\nu}$ is the small perturbation to this metric.

4.1 Perturbed Curvature Quantities

(*For more details check out section 5.1 from Cosmology by Steven Weinberg, 2008)
The perturbation in the Christoffel coefficients of the second kind ($\Gamma^{\mu}_{\nu\lambda}$) will be given by:

$$\delta\Gamma^{\mu}_{\nu\lambda} = \frac{1}{2}\,\bar{g}^{\mu\rho}\left[-2h_{\rho\sigma}\,\bar{\Gamma}^{\sigma}_{\nu\lambda} + \partial_\lambda h_{\rho\nu} + \partial_\nu h_{\rho\lambda} - \partial_\rho h_{\lambda\nu}\right]$$

In Ricci Tensor ($R_{\mu\nu}$) it'll be given by:

$$\delta R_{\mu\kappa} = \frac{\partial\,\delta\Gamma^{\lambda}_{\mu\lambda}}{\partial x^{\kappa}} - \frac{\partial\,\delta\Gamma^{\lambda}_{\mu\kappa}}{\partial x^{\lambda}} + \delta\Gamma^{\nu}_{\mu\nu}\,\bar{\Gamma}^{\nu}_{\kappa\eta} + \delta\Gamma^{\nu}_{\kappa\eta}\,\bar{\Gamma}^{\eta}_{\mu\nu} - \delta\Gamma^{\nu}_{\mu\kappa}\,\bar{\Gamma}^{\eta}_{\nu\eta} - \delta\Gamma^{\nu}_{\nu\eta}\,\bar{\Gamma}^{\eta}_{\mu\kappa}$$

In the Einstein equation (10) the quantity in the brackets is called the Source Tensor ($S_{\mu\nu}$)

$$R_{\mu\nu} = -8\pi G(T_{\mu\nu} - 1/2g_{\mu\nu}T) = -8\pi G S_{\mu\nu} \Rightarrow S_{\mu\nu} = T_{\mu\nu} - 1/2g_{\mu\nu}T \tag{16}$$

and the perturbations in this source tensor is given by:

$$\delta S_{\mu\nu} = \delta T_{\mu\nu} - \frac{1}{2}\,\bar{g}_{\mu\nu}\,\delta T^{\lambda}_{\lambda} - \frac{1}{2}\,h_{\mu\nu}\,\bar{T}^{\lambda}_{\lambda}$$

(*Here, due to homogeneity and isotropy of the unperturbed energy-momentum tensor $\bar{T}_{\mu\nu}$, it'll take the form of that of a perfect fluid, and $\bar{u}^{\mu} = (1, 0, 0, 0)$ i.e. we are in the comoving frame.)

Now, we can construct the perturbed Einstein's equations using equation (16) as:

$$\delta R_{\mu\nu} = -8\pi G \delta S_{\mu\nu}$$

And lastly the conservation equation ($T^{\mu}_{\nu;\mu}$) to first order in perturbations shall give:

$$\partial_\mu \delta T^{\mu}_{\nu} + \bar{\Gamma}^{\mu}_{\mu\lambda}\,\delta T^{\lambda}_{\nu} - \bar{\Gamma}^{\lambda}_{\mu\nu}\,\delta T^{\mu}_{\lambda} + \delta\Gamma^{\mu}_{\mu\lambda}\,\bar{T}^{\lambda}_{\nu} - \delta\Gamma^{\lambda}_{\mu\nu}\,\bar{T}^{\mu}_{\lambda} = 0$$

Putting $\nu = j$ will give us the momentum conservation equation and putting $\nu = 0$ will give us the energy conservation equation.

5 SVT Decomposition

The spatial isotropy and homogeneity of the unperturbed metric ($\bar{g}_{\mu\nu}$) and the energy momentum tensor ($\bar{T}_{\mu\nu}$) allows decomposing their perturbations into scalar, divergenceless vector and divergenceless and traceless tensor modes which do not couple with each other upto the **first order**.

The spatial isotropy and homogeneity means that the background spatial slices have full SO(3) rotational symmetry. And, hence we can do the SVT decomposition.

The metric perturbation $h_{\mu\nu}$ can be put in the following form:

$$
\begin{aligned}
h_{00} &= -E, \\
h_{i0} &= a\left[\frac{\partial F}{\partial x^i} + G_i\right], \\
h_{ij} &= a^2\left[A\delta_{ij} + \frac{\partial^2 B}{\partial x^i \partial x^j} + \frac{\partial C_i}{\partial x^j} + \frac{\partial C_j}{\partial x^i} + D_{ij}\right]
\end{aligned}
$$

where A, B, C_i, $D_{ij} = D_{ji}$, E, F, G_i are function of $\mathbf{x}$ and t and satisfy:

$$
\frac{\partial C_i}{\partial x^i} = \frac{\partial G_i}{\partial x^i} = 0, \qquad \frac{\partial D_{ij}}{\partial x^i} = 0, \qquad D_{ii} = 0
$$

Similarly we can do an SVT decomposition of the perturbations of $T_{\mu\nu}$:

$$
\begin{aligned}
\delta T_{00} &= -\bar{\rho}\, h_{00} + \delta\rho, \\
\delta T_{i0} &= \bar{p}\, h_{i0} - (\bar{\rho} + \bar{p})\left(\partial_i \delta u + \delta u_i^V\right), \\
\delta T_{ij} &= \bar{p}\, h_{ij} + a^2\left[\delta_{ij}\,\delta p + \partial_i\partial_j \pi^S + \partial_i \pi_j^V + \partial_j \pi_i^V + \pi_{ij}^T\right]
\end{aligned}
$$

where, π_i^v, π_{ij}^T, δu_i^V satisfy the following conditions:

$$
\partial_i \pi_i^V = \partial_i \delta u_i^V = 0, \qquad \partial_i \pi_{ij}^T = 0, \qquad \pi_{ii}^T = 0
$$

Also, here we can show by writing $\delta u_0 = \delta(g_{0\nu} u^\nu)$ and expanding the RHS by the chain rule, omitting the second order terms and using the values of $\bar{u}^\mu, \bar{g}_{\mu\nu}$ that $\delta u_0 + \delta u^0 = h_{00}$. Now, by opening the normalization condition $g_{\mu\nu} u^\mu u^\nu = -1$ and keeping only the first order terms we can say that $\delta u_0 = \delta u^0 = h_{00}/2$.

The perturbations in the mixed energy-momentum tensor T^μ_ν shall be:

$$
\begin{aligned}
\delta T^i{}_j &= \delta_{ij}\,\delta p + \partial_i\partial_j \pi^S + \partial_i \pi_j^V + \partial_j \pi_i^V + \pi_{ij}^T, \\
\delta T^i{}_0 &= a^{-2}(\bar{\rho} + \bar{p})\left(a\,\partial_i F + a\,G_i - \partial_i \delta u - \delta u_i^V\right), \\
\delta T^0{}_i &= (\bar{\rho} + \bar{p})\left(\partial_i \delta u + \delta u_i^V\right), \qquad \delta T^0{}_0 = -\delta\rho, \\
\delta T^\lambda{}_\lambda &= 3\,\delta p - \delta\rho + \nabla^2 \pi^S
\end{aligned}
$$

5.1 Vector Modes

We get the following equations after running the code for vector perturbations,
Momentum Conservation Equation:

$$a[\bar{p}\partial_t G_i - HG_i(\bar{\rho} - 3\bar{p}) + G_i\partial_t\bar{p}] - \bar{p}\nabla^2 C_i + \{3H(\bar{p} + \bar{\rho})\delta U_i^V + \partial_t[(\bar{p} + \bar{\rho})\delta U_i^V]\} = 0 \quad (17)$$

Einstein 0i Equation:

$$\frac{1}{2a}\{4G_i\dot{a}^2 + 2G_i a\ddot{a} + 8\pi\bar{G}a^2(\bar{p} - \bar{\rho})G_i - 16\pi\bar{G}(\bar{p} + \bar{\rho})u_i^V + a\partial_t\nabla^2 C_i - \nabla^2 G_i\} = 0 \quad (18)$$

Einstein ij Equation:

$$\begin{aligned} &\frac{a^2}{2}\partial_t^2 q_{ij} + \frac{3a\dot{a}}{2}\partial_t q_{ij} + [4\pi\bar{G}a^2(\bar{p} - \bar{\rho}) + 2\dot{a}^2 + a\ddot{a}]q_{ij} - \frac{a}{2}\partial_t l_{ij} - \dot{a}l_{ij} = 0 \\ &, l_{ij} = \partial_i G_j + \partial_j G_i \ \ , q_{ij} = \partial_i C_j + \partial_j C_i \end{aligned} \quad (19)$$

In vector mode only G_i, C_i, π_i^V, $u_i^V = \delta u_i^V$ (since, $\bar{u}_i^V = 0$) survive.
(*Since we have a perfect fluid therefore $\pi_i^V = 0$)

5.1.1 Gauge Invariant Quantities

*(Look at appendix B to know the effect of gauge on $h_\mu\nu$)
We'll find the gauge invariant quantities for this mode, since these quantities shall be physically relevant and then write our equations in terms of these quantities.

First of all our gauge ζ^μ can be Helmholtz decomposed:

$$\begin{aligned} &\zeta^\mu = (\zeta^0, \zeta^i) = (\zeta^0, \partial^i\zeta + \zeta_T^i), \ \ \partial_i\zeta_T^i = 0, \\ &\zeta \text{ is the solution to the equation } \partial^i\partial_i\zeta = \partial_i\zeta^i \end{aligned}$$

Now, since we are dealing with pure vector perturbations therefore all the scalar parts shall be zero, i.e. here $\zeta^0 = 0, \partial^i\zeta = 0$, hence:

$$\begin{aligned} &\zeta^\mu = (0, \zeta_T^i) \\ &\Rightarrow \zeta^i = \zeta_T^i \Rightarrow \partial_i\zeta^i = \partial_i\zeta_T^i = 0 \Rightarrow \partial_i\zeta^i = 0 \\ &\Rightarrow \zeta^\mu = (0, \zeta^i), \ \ \partial_i\zeta^i = 0 \end{aligned}$$

Now, we shall find the change in G_i introduced by the gauge.

We know that h_{0i} in vector mode shall be, $h_{0i} = aG_i$
$\Rightarrow \delta h_{0i} = a\delta G_i$
From equation (B.4) we can say that $\delta h_{0i} = -\nabla_0\zeta_i - \nabla_i\zeta_0$
$\Rightarrow$ Since, $\zeta_0 = 0$ hence, $\delta h_{0i} = -\nabla_0\zeta_i = -(\partial_t\zeta_i - \Gamma_{0i}^\lambda\zeta_\lambda)$
$\Rightarrow \delta h_{0i} = -(\partial_t\zeta_i - \Gamma_{0i}^\lambda\zeta_\lambda) = -(\partial_t\zeta_i - \bar{\Gamma}_{0i}^\lambda\zeta_\lambda)$ (Omitting the 2nd order term $\delta\Gamma_{0i}^\lambda\zeta_\lambda$)
$\Rightarrow \delta h_{0i} = -(\partial_t\zeta_i - \frac{\dot{a}}{a}\zeta_\lambda) = -(\partial_t\zeta_i - H\zeta_\lambda)$
$\Rightarrow$ Hence, $a\delta G_i = -(\partial_t\zeta_i - H\zeta_\lambda)$

$$\Rightarrow \delta G_i = \frac{-1}{a}(\partial_t \zeta_i - H\zeta_\lambda) \tag{20}$$

Now, for change in C_i due to gauge:

Since, $h_i j = a^2(\partial_j C_i + \partial_i C_j) \Rightarrow \delta h_i j = a^2(\partial_j \delta C_i + \partial_i \delta C_j)$

From equation (B.4) $\delta h_{ij} = -\nabla_i \zeta_j - \nabla_j \zeta_i = -(\partial_i \zeta_i - \Gamma^\lambda_{ij}\zeta_\lambda) - (\partial_j \zeta_i - \Gamma^\lambda_{ji}\zeta_\lambda)$

$\Rightarrow \delta h_{ij} = -\partial_i \zeta_i - \partial_j \zeta_i + 2\Gamma^\lambda_{ij}\zeta_\lambda = -\partial_i \zeta_i - \partial_j \zeta_i + 2\bar{\Gamma}^\lambda_{ij}\zeta_\lambda$

$\Rightarrow \delta h_{ij} = -\partial_i \zeta_i - \partial_j \zeta_i + 2(\bar{\Gamma}^0_{ij}\zeta_0 + \bar{\Gamma}^k_{ij}\zeta_k) = -\partial_i \zeta_i - \partial_j \zeta_i + 2\bar{\Gamma}^0_{ij}\zeta_0 = -\partial_i \zeta_i - \partial_j \zeta_i + 2a^2 H \delta_{ij}\zeta_0$

$\Rightarrow \delta h_{ij} = -\partial_i \zeta_i - \partial_j \zeta_i$

$\Rightarrow$ Hence, $a^2(\partial_j C_i + \partial_i C_j) = -\partial_i \zeta_i - \partial_j \zeta_i$

$$\Rightarrow \delta C_i = -\frac{\zeta_i}{a^2} \tag{21}$$

It can be shown that the quantity $V_i = G_i - a\dot{C}_i$ is gauge invariant (i.e. $\delta G_i = 0$). (22)

5.1.2 Solving the Conservation and Einstein's Equations

From equation (19), inserting the values of q_{ij} and l_{ij} and taking their time derivatives and rearranging a bit we'll get the following equation:

$$\partial_i(\dot{V}_j + 2HV_j) + \partial_j(\dot{V}_i + 2HV_i) = 0$$

Let the quantity $\dot{V}_j + 2HV_j$ be called W_j therefore we'll get:

$\partial_i(W_j) + \partial_j(W_i) = 0$

$\Rightarrow \partial_i\partial_i(W_j) + \partial_i\partial_j(W_i) = 0 \Rightarrow \nabla^2 W_j + \partial_j(\partial_i W_i) = 0$

here, $\partial_i W_i = \partial_i \dot{V}_i + 2H\partial_i V_i = \partial_t(\partial_i V_i) = 0$, ($\partial_i V_i = 0$, since gauge invariant.)

$\Rightarrow \nabla^2 W_j = 0 \Rightarrow W_j = 0 \Rightarrow \dot{V}_j + 2HV_j = 0$

Solving the differential equation we'll get

$V_j = \dfrac{k}{a^2}$, Where k is the integration constant

$$\Rightarrow G_j - a\dot{C}_j \propto \frac{1}{a^2} \tag{23}$$

From equation (18), using the Friedmann (equation (12)) and the Raychaudari (equation (11)) equations we'll get the following equation:

$$\nabla^2 V_i + 16\pi\bar{G}(\bar{p} + \bar{\rho})\delta u_i^V = 0$$

$$\Rightarrow \nabla^2 V_i \propto (\bar{p} + \bar{\rho})\delta u_i^V \tag{24}$$

Now, the gauge choice is arbitrary therefore we can take such a gauge choice where C_i becomes zero (i.e. $\delta C_i = C_i - \zeta_i/a^2 = 0 \Rightarrow$ if $\zeta_i = a^2 C_i$) therefore,

G_i will become $G_i - \dfrac{1}{a}(\dot{\zeta} - H\zeta) = G_i - \partial_t(aC_i) = G_i$ i.e. G_i is gauge invariant.

Therefore, from equation (23) we'll get $G_j \propto 1/a^2$ hence $G_i \to 0$.

Now, in equation (17), the momentum conservation equation we can observe that only the term in the curly braces on the right will survive cause $G_i, C_i = 0$ and hence we'll get:

$$3H(\bar{p}+\bar{\rho})\delta U_i^V + \partial_t[(\bar{p}+\bar{\rho})\delta U_i^V] = 0$$

Let $(\bar{p}+\bar{\rho})\delta U_i^V = X_i$ therefore,

$$3HX_i + \dot{X}_i = 0$$

Solving the equation we'll get:

$$X_i = (\bar{p}+\bar{\rho})\delta U_i^V \propto \frac{1}{a^3} \tag{25}$$

We can see that overall the quantities V_i and X_i decay with time as the universe expands.

5.2 Scalar Modes

We get the following equation after running the code for scalar mode.
Energy Conservation Equation:

$$\begin{aligned}&\partial_t(E\bar{p}) - \partial_t(\delta\rho) - H\bar{p}(\nabla^2 B) + \frac{2\bar{p}(\nabla^2 F)}{a} - H(\nabla^2\pi^S) + \frac{3\dot{A}(p+\rho)}{2} - \frac{(p+\rho)\partial_t(\nabla^2 B)}{2}\\ &- 3H[\bar{p}(A-E) + \delta p + \delta\rho] + \frac{(\bar{p}+\bar{\rho})\nabla^2 u^S}{a^2} = 0\end{aligned} \tag{26}$$

Momentum Conservation Equation:

$$\begin{aligned}&a\bar{p}\partial_t\partial_i F - \bar{p}\partial_i A - \bar{p}\partial_i(\nabla^2 B) - \frac{(p+\rho)\partial_i E}{2} + \partial_i F[a\dot{\bar{p}} - \dot{a}(p - 4\bar{p} + \rho)]\\ &-\partial_i(\delta p) - \partial_i(\nabla^2\pi^S) + (\bar{p}+\bar{\rho})\partial_t\partial_i u^S + [3(\bar{p}+\bar{\rho})H + (\dot{\bar{p}}+\dot{\bar{\rho}})]\partial_i u^S = 0\end{aligned} \tag{27}$$

Einstein 00 equation:

$$\begin{aligned}&-4\pi\bar{G}[-3\bar{p} + \bar{p}(\nabla^2 B) + 3\delta p + \delta\rho + \nabla^2\pi^S + 2(\bar{p}-\bar{\rho})E] + \frac{3\ddot{A}}{2} + 3H\dot{A} +\\ &\frac{\partial_t^2(\nabla^2 B)}{2} + H\partial_t(\nabla^2 B) - \frac{3H\dot{E}}{2} - \nabla^2 E - \frac{\partial_t(\nabla^2 F)}{a} - \frac{H(\nabla^2 F)}{a} = 0\end{aligned} \tag{28}$$

Einstein 0i equation:

$$\partial_t\partial_i A - [4\pi\bar{G}a(\bar{p}-\bar{\rho}) + \frac{2H}{a} + \ddot{a}]\partial_i F - H\partial_i E + 8\pi\bar{G}(\bar{p}+\bar{\rho})\partial_i u^S = 0 \tag{29}$$

Einstein ij Equation:

$$\begin{aligned}&\partial_i\partial_j\{-8\pi\bar{G}a^2\pi^S + \frac{a^2\ddot{B}}{2} + \frac{3a\dot{a}\dot{B}}{2} + B[4\pi\bar{G}a^2(\bar{p}-\bar{\rho}) + 2\dot{a}^2 + a\ddot{a}]\\ &- a\dot{F} - 2\dot{a}F - \frac{(A+E)}{2}\} = 0, i \neq j\end{aligned} \tag{30}$$

Einstein ii Equation:

$$4\pi\bar{G}a^2[\bar{p}E+\bar{p}(\nabla^2B-\partial_i^2B)+(\delta p-\delta\rho)+(\nabla^2\pi^S-2\partial_i^2\pi^S)]+\frac{a^2\ddot{A}}{2}+\frac{a^2\partial_t^2\partial_i^2B}{2}+3a\dot{a}\dot{A}+$$
$$a\dot{a}\partial_t[\frac{\nabla^2B}{2}+\frac{3\partial_i^2B}{2}]-\frac{a\dot{a}\dot{E}}{2}+2(A-E)\dot{a}^2+4\pi\bar{G}a^2(4\bar{p}-\bar{\rho})A+$$
$$\partial_i^2B[4\pi\bar{G}a^2(2\bar{p}-\bar{\rho})+2\dot{a}^2+a\ddot{a}]-a\partial_t\partial_i^2F-\dot{a}(\nabla^2F+2\partial_i^2F)+a\ddot{a}(A-E)-\frac{(\nabla^2A+\partial_i^2A)}{2}$$
$$-\frac{\partial_i^2E}{2}=0 \tag{31}$$

In scalar mode only $A, B, E, F, \delta p, \delta\rho, \pi^S$ survive.

5.2.1 Gauge Invariant Quantities

First of all our gauge ζ^μ can be Helmholtz decomposed:

$$\zeta^\mu=(\zeta^0,\zeta^i)=(\zeta^0,\partial^i\zeta+\zeta_T^i),\ \ \partial_i\zeta_T^i=0,$$
$$\zeta \text{ is the solution to the equation } \partial^i\partial_i\zeta=\partial_i\zeta^i$$

Now, since we are dealing with pure scalar perturbations therefore all the vector parts shall be zero, i.e. here $\zeta_T^i=0$, hence:

$$\zeta^\mu=(\zeta^0,\partial^i\zeta)$$

Finding the change in E due to the gauge:

$$\text{We know that } h_{00}=-E \text{ i.e. } \delta h_{00}=-\delta E$$
$$\text{From equation (B.4) we can say that } \delta h_{00}=-2\nabla_0\zeta_0$$
$$\Rightarrow \delta h_{00}=-2(\partial_0\zeta_0-\bar{\Gamma}^\lambda_{00}\zeta_\lambda)=-2\dot{\zeta}_0$$
$$\Rightarrow \delta E=2\dot{\zeta}_0 \tag{32}$$

Finding the change in F due to the gauge:

$$\text{We know that in scalar mode } h_{0i}=a\partial_iF\Rightarrow\delta h_{0i}=a\partial_i\delta F$$
$$\text{From equation (B.4) we can say that } \delta h_{0i}=-\nabla_0\zeta_i-\nabla_i\zeta_0$$
$$\Rightarrow \delta h_{0i}=-(\partial_0\zeta_i-\bar{\Gamma}^\lambda_{0i}\zeta_\lambda)-(\partial_i\zeta_0-\bar{\Gamma}^\lambda_{i0}\zeta_\lambda)=-\dot{\zeta}-\partial_i\zeta_0+2H\zeta_i$$
$$\text{Since, } \zeta_i=\partial_i\zeta \text{ therefore } \delta h_{0i}=-\dot{\zeta}-\partial_i\zeta_0+2H\partial_i\zeta$$
$$\Rightarrow a\partial_i\delta F=-\dot{\zeta}-\partial_i\zeta_0+2H\partial_i\zeta$$
$$\delta F=\frac{-1}{a}(\dot{\zeta}+\zeta_0-2H\zeta) \tag{33}$$

Finding the change in A and B due to the gauge:

$$\text{Since, in scalar mode } h_{ij}=a^2(A\delta_{ij}+\partial_{ij}^2B)\Rightarrow\delta h_{ij}=a^2(\delta A\delta_{ij}+\partial_{ij}^3\delta B)$$
$$\text{From equation (B.4) } \delta h_{ij}=-\nabla_i\zeta_j-\nabla_j\zeta_i$$
$$\Rightarrow \delta h_{ij}=-(\partial_i\zeta_j-\bar{\Gamma}^\lambda_{ij}\zeta_\lambda)-(\partial_j\zeta_i-\bar{\Gamma}^\lambda_{ji}\zeta_\lambda)=-\partial_i\zeta_j-\partial_j\zeta_i+2\bar{\Gamma}^0_{ij}\zeta_0$$
$$\Rightarrow \delta h_{ij}=-\partial_i\zeta_j-\partial_j\zeta_i+2a^2H\zeta_0\delta_{ij}=-\partial_i\partial_j\zeta-\partial_j\partial_i\zeta+2a^2H\zeta_0\delta_{ij}$$
$$\Rightarrow a^2(\delta A\delta_{ij}+\partial_{ij}^3\delta B)=-2\partial_i\partial_j\zeta+2a^2H\zeta_0\delta_{ij}$$

$$\Rightarrow \delta A = 2H\zeta_0$$
$$\delta B = \frac{-2}{a^2}\zeta \tag{34}$$

We shall construct the quantity σ defined as:

$$\sigma = aF - \frac{a^2}{2}\dot{B}$$

We can also see that $\delta\sigma = -\zeta_0$.
Now, We can construct 2 gauge invariant quantities defined as:

$$\Psi = A + 2H\sigma = A + 2\dot{a}F - \dot{a}a\dot{B}$$
$$\Phi = \frac{E}{2} + \dot{\sigma} = \frac{E}{2} + \dot{a}F + a\dot{F} - \frac{a^2\ddot{B}}{2} - a\dot{a}\dot{B} \tag{35}$$

A Symmetries and Killing Vectors

A.1 Symmetries

We have the **Geodesic equation**

$$\frac{d^2x^\mu}{d\tau^2} + \Gamma^\mu_{\alpha\beta}\frac{dx^\alpha}{d\tau}\frac{dx^\beta}{d\tau} = 0 \tag{A.1}$$

We can let $\frac{dx^\mu}{d\tau}$ be the the 4-velocity vector u^μ and hence our geodesic equation become

$$\frac{du^\mu}{d\tau} + \Gamma^\mu_{\alpha\beta}u^\alpha u^\beta = 0$$

Now since,

$$\begin{aligned}
&\nabla_\lambda V^\mu = \partial_\lambda V^\mu + \Gamma^\mu_{\lambda\nu}V^\nu \\
\Rightarrow\ &\nabla_\lambda u^\mu = \partial_\lambda u^\mu + \Gamma^\mu_{\lambda\nu}u^\nu \\
\Rightarrow\ &u^\lambda\nabla_\lambda u^\mu = u^\lambda\partial_\lambda u^\mu + u^\lambda\Gamma^\mu_{\lambda\nu}u^\nu \\
\Rightarrow\ &u^\lambda\nabla_\lambda u^\mu = \frac{du^\mu}{d\tau} + u^\lambda\Gamma^\mu_{\lambda\nu}u^\nu
\end{aligned}$$

and from the geodesic equation we will get $\frac{du^\mu}{d\tau} = -\Gamma^\mu_{\alpha\beta}u^\alpha u^\beta$ from which we can say that

$$u^\lambda\nabla_\lambda u^\mu = 0$$

further, since $p^\mu = mu^\mu$, therefore

$$p^\lambda\nabla_\lambda p^\mu = 0 \tag{A.2}$$

Now, since $\nabla_\lambda g_{\mu\nu} = 0$ therefore from $p^\lambda\nabla_\lambda p^\mu = 0$:

$$\begin{aligned}
&p^\lambda\nabla_\lambda g^{\mu\nu}p_\nu = p^\lambda g^{\mu\nu}\nabla_\lambda p_\nu = 0 \\
\Rightarrow\ &p^\lambda\nabla_\lambda p_\nu = p^\lambda\nabla_\lambda p_\mu = p^\lambda(\partial_\lambda p_\mu - \Gamma^\sigma_{\lambda\mu}p_\sigma) = 0 \\
\Rightarrow\ &p^\lambda\partial_\lambda p_\mu - \Gamma^\sigma_{\lambda\mu}p^\lambda p_\sigma = 0
\end{aligned}$$

The first term $p^\lambda\partial_\lambda p_\mu$ can be written as:

$$m\frac{dx^\lambda}{d\tau}\partial_\lambda p_\mu = m\frac{p_\mu}{d\tau}$$

and the second term $\Gamma^\sigma_{\lambda\mu}p^\lambda p_\sigma$ can be written as:

$$\begin{aligned}
&\frac{1}{2}g^{\sigma\nu}(\partial_\lambda g_{\mu\nu} + \partial_\mu g_{\nu\lambda} - \partial_\nu g_{\lambda\mu})p^\lambda p_\sigma \\
=&\frac{1}{2}[(\partial_\lambda g_{\mu\nu})p^\lambda p^\nu + (\partial_\mu g_{\nu\lambda})p^\lambda p^\nu - (\partial_\nu g_{\lambda\mu})p^\lambda p^\nu] \\
=&\frac{1}{2}(\partial_\mu g_{\nu\lambda})p^\lambda p^\nu
\end{aligned}$$

Therefore,

$$m\frac{p_\mu}{d\tau} = \frac{1}{2}(\partial_\mu g_{\nu\lambda})p^\lambda p^\nu \tag{A.3}$$

which is just another way to write the geodesic equation. Now, see that if for some x^σ, $\partial_\sigma g_{\mu\nu} = 0$ i.e. $g_{\mu\nu}$ is independent of x_σ, we can see that $\frac{dp^\sigma}{d\tau} = 0$ i.e. p_σ is constant.

This looks like that if $g_{\mu\nu}$ is independent of x_σ then p_σ is a constant but, the converse of this statement is not necessarily true. But, we can go to some other coordinate system where $g_{\mu\nu}$ is dependent on every other coordinate and there is no such x_σ such that $\partial_\sigma g_{\mu\nu} = 0$. This happens because symmetry is not coordinate based and rather geometry based. Hence a better way has to be developed to find the transforming over which p_σ will be constant.

A.2 Symmetries and Killing Vectors

Let $K = \partial_\sigma \Rightarrow K^\mu = \partial^\mu_\sigma = \delta^\mu_\sigma$. Hence, we can write p_σ as $K_\nu p^\nu$ or $K^\nu p_\nu$. Since, p_σ is a constant therefore

$$\frac{dp_\sigma}{d\tau} = \frac{dx^\mu}{d\tau}\frac{dp_\sigma}{dx^\mu} = \frac{dx^\mu}{d\tau}\nabla_\mu p^\sigma = p^\mu \nabla_\mu p^\sigma = 0$$

Therefore

$$\begin{aligned} & p^\mu \nabla_\mu p^\sigma = p^\mu \nabla_\mu (K_\nu p^\nu) = 0 \\ \Rightarrow\ & p^\mu K_\nu \nabla_\mu (p^\nu) + p^\mu p^\nu \nabla_\mu (K_\nu) = 0 \\ \Rightarrow\ & p^\mu p^\nu (\nabla_\mu K_\nu) = 0 \\ \Rightarrow\ & \nabla_\mu K_\nu = 0 \end{aligned}$$

Here, in the last step we used equation (8). Now,

$$\nabla_\mu K_\nu = \nabla_{(\mu} K_{\nu)} + \nabla_{[\mu} K_{\nu]}$$

where the first term is the symmetric part and second term is the antisymmetric part.

$$p^\mu p^\nu \nabla_\mu K_\nu = p^\mu p^\nu \nabla_{(\mu} K_{\nu)} + p^\mu p^\nu \nabla_{[\mu} K_{\nu]}$$

and since $p^\mu p^\nu \nabla_\mu K_\nu = p^\nu p^\mu \nabla_\nu K_\mu$ therefore

$$\begin{aligned} & p^\mu p^\nu \nabla_\mu K_\nu = p^\nu p^\mu \nabla_\nu K_\mu \\ \Rightarrow\ & p^\mu p^\nu \nabla_{(\mu} K_{\nu)} + p^\mu p^\nu \nabla_{[\mu} K_{\nu]} = p^\nu p^\mu \nabla_{(\nu} K_{\mu)} - p^\nu p^\mu \nabla_{[\nu} K_{\mu]} \\ \Rightarrow\ & p^\mu p^\nu \nabla_{[\mu} K_{\nu]} = 0 \end{aligned}$$

and hence

$$\nabla_{(\mu} K_{\nu)} = 0 \tag{A.4}$$

which is the **Killing's Equation**. Now, vector fields satisfying the Killing's Equation are called the **Killing Vectors**. These vector fields generate isometries, transforming under which the metric $g_{\mu\nu}$ remains invariant.

B Effects of gauge

Consider a transformation of our coordinates by a small ζ^μ upto the first order.

$$\begin{aligned}
&x^\mu \to x^{'\mu} = x^\mu + \zeta^\mu(x) \\
\Rightarrow\ &x^\mu = x^{'\mu} - \zeta^\mu(x) \\
\Rightarrow\ &x^\mu = x^{'\mu} - (\zeta^\mu(x') + (x^\alpha - x'^\alpha)\partial_\alpha\zeta^\mu(x') \ \ + \ \ ...) \ \text{(Taylor Expansion)} \\
\Rightarrow\ &x^\mu \simeq x^{'\mu} - \zeta^\mu(x')
\end{aligned}$$

Hence by using the above result,

$$\frac{\partial x^\alpha}{\partial x'^\mu} = \delta^\alpha_\mu - \partial_\mu\zeta^\alpha \tag{B.1}$$

And the perturbed metric in the x'^μ coordinates $g'_{\mu\nu}(x')$ will be

$$g'_{\mu\nu}(x') = \frac{\partial x^\alpha}{\partial x'^\mu}\frac{\partial x^\beta}{\partial x'^\nu}g_{\alpha\beta}(x) \tag{B.2}$$

Now, using (16) and (17) and ignoring the higher order terms one can say that

$$g'_{\mu\nu}(x') \simeq g_{\mu\nu}(x) - g_{\mu\beta}(x)\partial_\nu\zeta^\beta - g_{\alpha\nu}(x)\partial_\mu\zeta^\alpha$$

Now, do a Taylor expansion on LHS about x and ultimately we'll get

$$\begin{aligned}
&LHS = g'_{\mu\nu}(x') \simeq g'_{\mu\nu}(x) + \zeta^\lambda\partial_\lambda g_{\mu\nu}(x) \\
\Rightarrow\ &g'_{\mu\nu}(x) \simeq g_{\mu\nu}(x) - g_{\mu\beta}(x)\partial_\nu\zeta^\beta - g_{\alpha\nu}(x)\partial_\mu\zeta^\alpha - \zeta^\lambda\partial_\lambda g_{\mu\nu} \\
\Rightarrow\ &g'_{\mu\nu}(x) \simeq g_{\mu\nu}(x) - \nabla_\mu\zeta_\nu - \nabla_\nu\zeta_\mu
\end{aligned} \tag{B.3}$$

The third step can be proved by opening $\nabla_\mu\zeta_\nu + \nabla_\nu\zeta_\mu$ and rearranging the terms.

Now, g and g' can be written as

$$\begin{aligned}
g_{\mu\nu} &= \bar{g}_{\mu\nu} + h_{\mu\nu} \\
g'_{\mu\nu} &= \bar{g} + h'_{\mu\nu}
\end{aligned}$$

Since, $\bar{g}_{\mu\nu}$ and $\bar{g}$ are the background FRW metrics, due to homogeneity and isotropy of the background space they should be the same, and hence

$$h_{\mu\nu}(x) \to h'_{\mu\nu}(x) = h_{\mu\nu}(x) - \nabla_\mu\zeta_\nu - \nabla_\nu\zeta_\mu \tag{B.4}$$

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